

# LAB SHEET-3

## Objectives

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

## DDL Questions (Data Definition Language)

- Create the **TEACHER** table with the following attributes:  
(Name, DeptNo, Date\_of\_Joining, DeptName, Location, Salary).

```
1  ● ⊖ CREATE TABLE TEACHER (  
2      Name VARCHAR2(50),  
3      DeptNo NUMBER,  
4      Date_of_Joining DATE,  
5      DeptName VARCHAR2(50),  
6      Location VARCHAR2(50),  
7      Salary NUMBER  
8  ) ;
```

- Create the **SALES** table with the following attributes:  
(Sales\_No, Salesname, Branch, Salesamount, DOB).

```
1  ● ⊖ CREATE TABLE SALES (  
2      Sales_No NUMBER,  
3      Salesname VARCHAR2(50),  
4      Branch VARCHAR2(50),  
5      Salesamount NUMBER,  
6      DOB DATE  
7  ) ;
```

- Create the **DEPARTMENTS** table with the following attributes:  
(dept\_no, dept\_name, dept\_location).

```
44  ● ⊖ CREATE TABLE DEPARTMENTS (  
45  ✖      dept_no NUMBER PRIMARY KEY,  
46      dept_name VARCHAR2(50),  
47      dept_location VARCHAR2(50)  
48  ) ;
```

- Create the **EMPLOYEES** table with the following attributes:  
(emp\_id, emp\_name, emp\_salary, dept\_no).

```

55  CREATE TABLE EMPLOYEES (
56      emp_id NUMBER PRIMARY KEY,
57      emp_name VARCHAR2(50),
58      emp_salary NUMBER,
59      dept_no NUMBER
60  );

```

## DML Questions (Data Manipulation Language)

### From TEACHER Table

- Insert five records into the **TEACHER** table.

```

11  INSERT INTO TEACHER VALUES
12      ('Aman',1,DATE '2020-01-01','Mathematics','Delhi',40000);
13  INSERT INTO TEACHER VALUES
14      ('Riya',2,DATE '2019-05-10','Commerce','Noida',35000);
15  INSERT INTO TEACHER VALUES
16      ('Neha',1,DATE '2018-03-15','Mathematics','Delhi',42000);
17  INSERT INTO TEACHER VALUES
18      ('Arjun',3,DATE '2021-07-01','Science','Gurgaon',30000);
19  INSERT INTO TEACHER VALUES
20      ('Pooja',2,DATE '2017-09-12','Commerce','Noida',38000);

```

Give an increment of 25% salary to teachers working in the Mathematics department.

```

23  UPDATE TEACHER
24      SET Salary = Salary * 1.25
25      WHERE DeptName = 'Mathematics';

```

- Give an increment of 15% salary to teachers working in the Commerce department.

```

29  UPDATE TEACHER
30      SET Salary = Salary * 1.15
31      WHERE DeptName = 'Commerce';

```

## From SALES Table

- Insert five records into the **SALES** table.

```
65 • INSERT INTO SALES VALUES (1, 'Rahul', 'Delhi', 5000, DATE '1998-12-21');
66 • INSERT INTO SALES VALUES (2, 'Aman', 'Mumbai', 7000, DATE '1997-05-10');
67 • INSERT INTO SALES VALUES (3, 'Neha', 'Delhi', 6000, DATE '1999-12-15');
68 • INSERT INTO SALES VALUES (4, 'Riya', 'Pune', 4000, DATE '1996-08-20');
69 • INSERT INTO SALES VALUES (5, 'Arjun', 'Mumbai', 8000, DATE '1995-12-05');
```

- Calculate the total sales amount in each branch.

```
17 • SELECT Branch, SUM(Salesamount) AS Total_Sales
18 FROM SALES
19 GROUP BY Branch;
```

- Calculate the average sales amount in each branch.

```
22 • SELECT Branch, AVG(Salesamount) AS Average_Sales
23 FROM SALES
24 GROUP BY Branch;
```

- Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.

```
28 • SELECT Salesname,
29 TO_CHAR(DOB, 'DD-Mon-YY') AS DOB
30 FROM SALES
31 WHERE TO_CHAR(DOB, 'MM')='12';
32
```

- Display the Salesname and DOB of salesmen in alphabetical order of the month.

```
36 • SELECT Salesname, DOB
37 FROM SALES
38 ORDER BY TO_CHAR(DOB, 'Month');
39
```

## From JOIN Operations

- Display employee details and department details where the department numbers are the same in both tables.

```
1 • INSERT INTO DEPARTMENTS VALUES (1,'HR','Delhi');
2 • INSERT INTO DEPARTMENTS VALUES (2,'Finance','Mumbai');
3 • INSERT INTO DEPARTMENTS VALUES (3,'IT','Noida');
4
5 • INSERT INTO EMPLOYEES VALUES (101,'Aman',50000,1);
6 • INSERT INTO EMPLOYEES VALUES (102,'Riya',60000,2);
7 • INSERT INTO EMPLOYEES VALUES (103,'Neha',55000,3);
8 • INSERT INTO EMPLOYEES VALUES (104,'Arjun',45000,2);
```

```
11 • SELECT *
12     FROM EMPLOYEES e
13     JOIN DEPARTMENTS d
14     ON e.dept_no = d.dept_no;
```

- Display the employee name and department name using a LEFT OUTER JOIN.

```
16 • SELECT e.emp_name, d.dept_name
17     FROM EMPLOYEES e
18     LEFT JOIN DEPARTMENTS d
19     ON e.dept_no = d.dept_no;
```

- Display the employee name and department name using a RIGHT OUTER JOIN.

```
22 • SELECT e.emp_name, d.dept_name
23     FROM EMPLOYEES e
24     RIGHT JOIN DEPARTMENTS d
25     ON e.dept_no = d.dept_no;
```

- Display the details of employees who draw a salary greater than the average salary.

```
29 • SELECT *
30     FROM EMPLOYEES
31     WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

- **TCL Questions (Transaction Control Language)**
- Perform the **ROLLBACK** command on the TEACHER table.
- Perform the **COMMIT** command on the TEACHER table.
- Write a query to implement a **SAVEPOINT**.

```
35 • SAVEPOINT A;
36 • ROLLBACK TO A;
37 • COMMIT;
```

### DCL Questions (Data Control Language)

- Write a query to **grant all privileges** on the EMPLOYEES table to DEPARTMENTS.

```
42 • GRANT ALL ON EMPLOYEES TO DEPARTMENTS;
```

- Write a query to **grant selected privileges** on the EMPLOYEES table to DEPARTMENTS.

```
44 • GRANT SELECT, INSERT ON EMPLOYEES TO DEPARTMENTS;
```

- Write a query to **revoke all privileges** on the EMPLOYEES table from DEPARTMENTS.
- Write a query to **revoke selected privileges** on the EMPLOYEES table from DEPARTMENTS.

```
50 • REVOKE SELECT, INSERT ON EMPLOYEES FROM DEPARTMENTS;
```